



# Data-Driven Problem Framing

Week 1 • Day 5 • Customer-Centric Foundations

Technical Product Manager Academy



## Day 5 Objectives

- Write a **problem statement** an engineer can scope from
- Build an **evidence brief** that pairs every claim with its tier and source
- Use AI to pressure-test — not author — the problem statement
- Present a **10-minute integrated Week 1 artifact** to the full cohort
- Set up the bridge to Week 2 (metrics + strategic design)



# Today's Shape

| Clock         | Block                     | Outcome                      |
|---------------|---------------------------|------------------------------|
| 09:00 – 09:15 | Opening                   | Review Week 1 artifact stack |
| 09:15 – 10:30 | Block 1 + Activity 1      | Problem statement template   |
| 10:45 – 12:00 | Block 2 + Activity 2      | Evidence brief & AI critique |
| 13:00 – 14:15 | Block 3 + Activity 3      | Rehearsal + peer dry-run     |
| 14:30 – 16:00 | Block 4 + Readouts + Wrap | Full-cohort presentations    |

# Block 1 — The Problem Statement

Scoped, verb-led, evidence-backed, falsifiable

# The Template

**Who** — [persona role + context]

**Currently** — [observed behavior, with frequency]

**Which causes** — [measurable consequence]

**Because** — [root cause, from 5 Whys]

**We know this because** — [evidence, tagged]

**Success looks like** — [the change we'd see if the problem is solved]

The last line is the one most problem statements skip. Without it, there's no way to know when you're done.



# A FieldPulse Example

**Who** — Lead Dispatchers at 50–200 tech HVAC companies

**Currently** — Spend an average of 45 minutes post-shift reconciling paper tickets, truck stock, and timecards (observed, 6 of 8 ride-alongs)

**Which causes** — ~3.75 unpaid hours per dispatcher per week and a measured 22% turnover in the role (reported, HR data from 3 customers)

**Because** — Ticket submission requires connectivity techs don't have at the last job, so paper becomes the path of least resistance (5 Whys root)

**We know this because** — **Observed** 6 ride-alongs; **Reported** 4 interviews;  
**Inferred** turnover causality

**Success looks like** — Post-shift reconcile under 10 minutes for 80% of dispatchers within 6 months

# Failure Modes of Problem Statements

| Failure              | Sounds like                                      | Why it's weak                            |
|----------------------|--------------------------------------------------|------------------------------------------|
| Solution in disguise | "Dispatchers need a mobile reconcile screen"     | Pre-decides the how; not falsifiable     |
| Consequence-free     | "Dispatchers are frustrated with reconciliation" | No measurable cost; no way to prioritize |
| No root cause        | "The process is inefficient"                     | Symptom, not mechanism                   |
| Ungrounded           | "Customers tell us..."                           | No evidence tier; can't be audited       |
| Success-agnostic     | Missing the last line                            | Can't recognize solved                   |

# Activity 1 — Draft Your Statement

Triads • 45 minutes

Using your Top 3 pain set and persona, draft a problem statement in the six-line template.

- Start with the strongest pain from Day 4
- Every line must survive the failure-mode table
- Mark which evidence tier each claim uses



20 min draft • 15 min failure-mode audit • 10 min peer-within-triad read



# Block 2 — The Evidence Brief

What goes with the problem statement



# The Brief, One Page

1. **The persona** — one paragraph from your Day 3 canvas
2. **The pain map** — your Day 4 matrix, with Top 3 highlighted
3. **The problem statement** — from Activity 1
4. **The evidence log** — each claim: source, date, tier, who captured it
5. **The open questions** — what you don't yet know and plan to learn

**The discipline:** Every claim in the problem statement appears in the evidence log. Nothing floats unsupported.



# AI as Pressure-Tester

Role: Senior engineering lead reading a problem statement I intend to scope against.  
Context: [paste problem statement + evidence log]  
Constraints:  
- Do not rewrite. Ask the questions you would ask me in a scoping meeting.  
- Rank your questions by how much they'd change your estimate if unanswered.  
- Flag any claim that is inferred or AI-generated that you'd want upgraded first.  
Format: Numbered list of questions, ranked.

**Rule:** AI can challenge the statement. AI does not get to write it.  
Ownership stays with the triad.

## Activity 2 — Pressure Test

Triads • 45 minutes

Run the pressure-test prompt. Then, as a triad:

- Answer every question you can from existing evidence
- Add unresolved questions to "Open questions" in the brief
- Upgrade or remove any claim that failed



10 min run prompt • 25 min respond • 10 min revise brief

# Block 3 — Rehearsal

10 minutes, the right 10 minutes



# The Readout Structure

1. **0:00 – 1:30** Persona (who, observed behavior, 1 quote)
2. **1:30 – 3:30** Top 3 pains (matrix, one slide)
3. **3:30 – 5:30** Problem statement (read it, one line at a time)
4. **5:30 – 7:30** Evidence sample + open questions
5. **7:30 – 10:00** Q&A from room, questions only

**Norm:** Don't defend in the Q&A. Listen and note. You'll act on the feedback in Week 2 pre-work.

## Activity 3 — Paired Dry-Run

Paired triads • 45 minutes

Triads pair. Each does a **full timed 10-minute readout** to the other, with stopwatch.

- Partner writes down the **3 questions** they would ask in real Q&A
- Partner writes down **1 sentence** of what they heard the problem to be
- Compare that sentence to what the triad intended to convey



10 min each direction • 25 min debrief

# Block 4 — Cohort Readouts

Week 1 integrated artifact





## How We Score (Reminder)

| Dimension                 | Weight |
|---------------------------|--------|
| Customer authenticity     | 25%    |
| Pain specificity          | 25%    |
| Evidence quality          | 20%    |
| Problem statement clarity | 20%    |
| AI-use transparency       | 10%    |

Each triad gets a score sheet from every peer triad + the instructor. Scores plus one strength and one suggestion.

# Week 1 Readouts

Full cohort • 10 min per triad + 3 min score-sheet writing

Order is random (draw triad names). Between triads, a 3-minute quiet block for written feedback.

- Presenting triad: follow the rehearsal structure
- Audience triads: questions only, no defending
- Everyone: written score + 1 strength + 1 suggestion



Plan ~80–90 min for 6 triads



# Week 1 Wrap

## What we built (five artifacts)

1. Prompt Pattern Library (triad)
2. PR/FAQ
3. Validated persona canvas
4. Ranked pain-point matrix
5. Problem statement + evidence brief

## What Week 2 does with them

- Problem statement → North Star metric
- Pain map → operational signals & KPIs
- Persona → customer journey map
- PR/FAQ → UX-principled re-framing

**Weekend pre-read for Week 2:** Re-read your problem statement. Bring 3 candidate metrics. Come curious about which one is the North Star.

# Closing Reflection

One sentence: what's the mindset you're taking into Week 2?